

The Origin of SARS-CoV-2 — Aletheia Case Seed

Suggested case ID: COV-001 · **Domains:** Virology · Epidemiology · Intelligence assessment · Congressional oversight · Research ethics · **Seed prepared:** 2026-08-25 · **Status convention:** *Supported / Contested / Unresolved / Contradicted* (per claim, per Aletheia convention)

Purpose of this document. This report is a comprehensive evidence seed for the Aletheia case on the origin of the COVID-19 pandemic. It decomposes the controversy into atomic claims, maps the evidence for and against each claim with exact provenance, covers the circumstantial record produced by recent congressional hearings and document releases (2023–2026, including the Fauci hearings), identifies where the disagreement actually lives, and proposes the tests that would settle each crux. It does not deliver a verdict: as of August 2026, no definitive evidence establishes either a natural spillover or a research-related incident, and every major institutional assessment carries explicit uncertainty qualifiers.

Executive Summary

Six years after SARS-CoV-2 emerged in Wuhan, the question of how it entered the human population remains formally unresolved, but the evidentiary landscape is no longer symmetric in the way it is often described. The **direct, positive evidence** — epidemiological clustering, phylodynamics, and market metagenomics — mostly points toward zoonotic spillover associated with the Huanan Seafood Wholesale Market: 66% of the first 41 hospitalized patients had direct market exposure, early cases without any market link lived significantly *closer* to the market than market-linked cases, and two early viral lineages are consistent with two spillover events in November–December 2019 ([Science](#) | [AAAS](#)) . The **circumstantial evidence** — proximity of China’s foremost SARS-coronavirus laboratory, a 2018 funding proposal to insert furin cleavage sites into bat coronaviruses in Wuhan, documented biosafety shortcomings, an offline sequence database, and a documented pattern of private-versus-public divergence among senior scientists and officials — points toward, or at least fails to exclude, a research-related incident ([archive.org](#)) .

The recent hearings and document releases have materially changed the record on **conduct**, not on **virology**. The 2024–2026 House and Senate investigations documented that senior NIAID adviser David Morens used private email to evade FOIA and described deleting federal records; that the authors of “The Proximal Origin of SARS-CoV-2” privately assigned non-trivial probabilities to a lab origin (Andersen 30%, Holmes 20%→10%) even while their paper said no laboratory scenario was plausible; and that Anthony Fauci, subpoenaed before the Senate on 29 July 2026, invoked the Fifth Amendment 111 times and was held in contempt of Congress on 6 August 2026, with a referral sent to the Department of Justice ([house.gov](#)) . None of this is evidence about where the virus came from — but it is direct evidence about the reliability of the early “consensus” narrative, which is itself one of the load-bearing claims in the case.

The decisive fact is structural: **both hypotheses currently rest partly on absences**. No infected intermediate animal has been identified despite roughly 80,000 animals tested in China; no SARS-CoV-2 progenitor has been identified in any laboratory collection; the WHO’s 27-member SAGO panel concluded in June 2025 that “the weight of available evidence ... suggests zoonotic spillover” while explicitly stating it could not even *assess* the laboratory hypothesis because China has not provided the underlying records ([Science](#) | [AAAS](#)) . The case is therefore genuinely open — and its resolution depends on a short list of specific, obtainable pieces of evidence enumerated in Section 9.

1. What is claimed

The case decomposes into two families of hypotheses, each with sub-variants that are frequently conflated in public argument. Precision here matters, because much of the six-year controversy has consisted of parties answering different claims.

Hypothesis family Z — zoonotic spillover (market-associated). SARS-CoV-2 evolved in bats (and possibly intermediate mammal hosts), reached Wuhan through the wildlife trade, and spilled into humans at or around the Huanan Seafood Wholesale Market in roughly November–December 2019, most likely in two independent introductions corresponding to lineages A and B ([Science](#) | [AAAS](#)) . This is the position of the mainstream virology literature (Worobey et al. and Pekar et al. in *Science*, July 2022; Crits-Christoph et al. in *Cell*, September 2024) and of the WHO’s SAGO panel, which in June 2025 found “the weight of available evidence” supports zoonotic spillover “either directly from bats or through an intermediate host,” while declining to rule anything else out ([Cell Press](#)) .

Hypothesis family L — research-related incident. SARS-CoV-2, or a close progenitor, was present in a Wuhan laboratory — most plausibly the Wuhan Institute of Virology (WIV) — through field collection or laboratory manipulation, and a researcher was infected and carried the virus into the city. Sub-variants range from *L1* (infection with an unmodified collected virus) through *L2* (infection with a virus altered by serial passage or reverse genetics, e.g. a furin-cleavage-site insertion of the type proposed in the 2018 DEFUSE bid) to *L3* (deliberate engineering/release), the last of which is rejected by the entire US intelligence community and by virtually all scientists on both sides ([FactCheck.org](#)) . This family is favored, with low-to-moderate confidence, by the FBI, the Department of Energy, the CIA (since January 2025), and Germany’s BND (since 2020, revealed 2025), and by the Republican majority of the House Select Subcommittee’s December 2024 final report ([ABC News](#)) .

Two framing claims sit underneath the origin question and are independently contested. The first is the “**consensus**” **claim** — that the scientific community converged on zoonosis through normal, data-driven deliberation. The second is the **suppression claim** — that senior US officials and funded scientists steered the early narrative away from a lab origin for reputational or institutional reasons. The 2024–2026 documentary record bears directly on these meta-claims, and they are where the hearings matter ([senate.gov](#)) .

Component claim (suggested ID)	Claim	Current status
COV-C001	The Huanan market was the early epicenter of the outbreak	Supported (epidemiology)
COV-C002	Two lineages imply 2 zoonotic introductions	Contested (phylodynamic inference)
COV-C003	Susceptible wildlife was present with virus at one stall	Supported (presence, not infection)
COV-C004	An infected intermediate host existed and can be found	Unresolved —none identified
COV-C005	No lab held SARS-CoV-2 or a close progenitor pre-outbreak	Unresolved —unverifiable while WIV records sealed
COV-C006	The furin cleavage site is anomalous among sarbecoviruses	Supported as fact; Contested as inference
COV-C007	DEFUSE-type work was proposed for Wuhan (documentary)	Supported as fact; relevance Contested

Table 1 – continued

Component claim (suggested ID)	Claim	Current status
COV-C008	WIV cultured SARS-like viruses at BSL-2	Supported (ODNI 2023)
COV-C009	WIV took its sequence database offline 12 Sep 2019	Supported as fact; motive Contested
COV-C010	WIV staff had COVID-like illness in autumn 2019	Unresolved (ODNI: “neither supports nor refutes”)
COV-C011	No verified circulation before ~Nov 2019	Supported (with data-access caveat)
COV-C012	Early public “consensus” diverged from private scientific views	Supported (Slack messages, diary, emails)

2. The stakes and the state of play

The pandemic killed on the order of **15 million people** in 2020–2021 by WHO excess-mortality estimation (14.83 million, uncertainty interval 13.2–16.6 million), roughly 2.7 times the officially reported death toll for that period, and reported deaths reached about 7.1 million by August 2025 — a figure widely considered a large undercount ([nih.gov](#)) . Knowing whether this began with a wildlife trader’s stall or a laboratory procedure determines which of two very different prevention regimes — wildlife-trade regulation or biosafety and research oversight — the world should prioritize, which is why the question remains institutionally alive rather than merely historical.

Public and institutional opinion have decoupled. By March 2023, American polls showed lab-leak belief at 64–66% versus 16–22% for natural origin (Quinnipiac; Economist/YouGov), even as the mainstream virology literature moved the *other* way ([The Washington Post](#)) . Within the US government there is no consensus: four intelligence entities plus the National Intelligence Council lean natural (low confidence), the FBI leans lab (moderate confidence), the Department of Energy and CIA lean lab (low confidence), and the ODNI’s June 2023 declassified report found “no direct evidence” of a biosafety incident while documenting the conditions that would make one plausible ([FactCheck.org](#)) . Internationally, Germany’s BND reportedly assessed a lab accident at 80–95% probability as early as 2020 — a conclusion the Merkel government did not publish — while China’s April 2025 white paper declares the lab leak “extremely unlikely,” points instead at the United States, and declares the investigation “finished” ([The Telegraph](#)) .

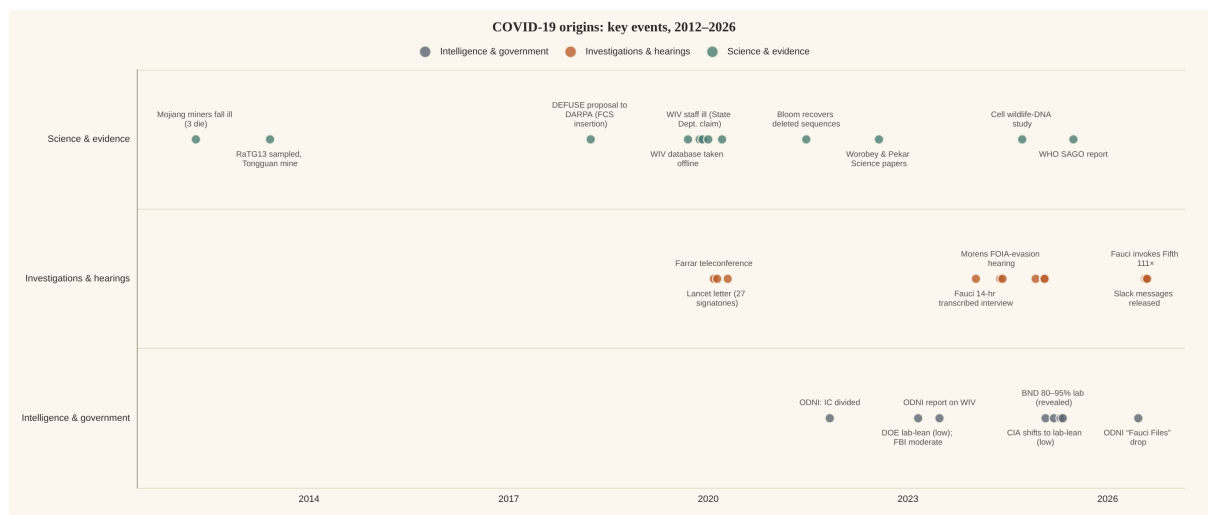


Figure 1: Timeline of key events, 2012–2026

Figure 1. Key events across three tracks: the scientific evidence base, the investigations and hearings, and intelligence/government actions. The 2024–2026 hearings cluster is the focus of Sections 5–6.

3. The scientific evidence for a market zoonosis

3.1 Early-case epidemiology points at the market

The earliest clinical description of the outbreak, Huang et al. in *The Lancet* (15 February 2020), covered 41 patients hospitalized by 2 January 2020, of whom **27 (66%) had direct exposure to the Huanan market**; the symptom onset of the first identified patient was 1 December 2019, and — a detail that has fueled six years of argument — “no epidemiological link was found between the first patient and later cases” (*The Lancet*) . The fuller WHO-mission case series found 55 of 168 early cases associated with the market, and the Worobey et al. reanalysis established the subtler pattern that carries the statistical weight: cases *without* any reported market link resided significantly **closer** to the market than cases who worked there, which is the signature expected if the market was the exposure site rather than an artifact of contact tracing, and excess pneumonia deaths rose first in the districts surrounding the market (*Science* | *AAAS*) .

Worobey and colleagues further showed that SARS-CoV-2-positive environmental samples inside the market were spatially concentrated in its western section, where live-mammal vendors operated, and that susceptible species — raccoon dogs, civets, bamboo rats and others — were being sold there in late 2019 (*Science* | *AAAS*) . Ascertainment bias has been the principal counterargument, and the paper answers it with several independent observations: all cases analyzed were hospitalized (so unlinked mild cases would not have been found by neighborhood searches); officials became aware of market-linked cases both near and far from the market simultaneously; and seroprevalence was later found highest in the districts around the market (*Science* | *AAAS*) . This remains the single strongest positive evidentiary thread in the entire case, and it is why the WHO SAGO panel, even while protesting Chinese non-cooperation, kept zoonosis in front (*WHO*) .

3.2 Two lineages, two introductions

Pekar et al. (*Science*, July 2022) analyzed early genomic diversity and found that pre-February-2020 SARS-CoV-2 comprised **two distinct lineages, A and B**, separated by two nucleotide differences, whose joint pattern is best explained by **at least two separate cross-species transmission events**: lineage B entering humans around 18 November 2019 (95% HPD 23 October–8 December) and lineage A within days to weeks afterward, with the lineage-B primary case preceding lineage A in 64.6% of posterior samples ([Science | AAAS](#)) . The authors’ simulations did not support a single introduction producing the observed pattern, and the analysis indicates the virus was unlikely to have circulated widely in humans before November 2019 — a bound that matters because it constrains both a cryptic pre-December community phase and a long pre-outbreak lab-incubation story ([Science | AAAS](#)) .

The two-introduction result is genuinely load-bearing for the zoonotic case — SAGO explicitly cited it as the pattern “rather than what would be expected from a lab leak — which would be a single virus introduction” ([World Health Organization \(WHO\)](#)) — and it is correspondingly contested. The inference depends on rooting choices and on the assumption that the two lineages reflect two jumps rather than one jump followed by early diversification and sampling gaps; the earliest lineage-A genomes lacked known market contact but came from people who lived or stayed close by, which both sides read in their favor ([Science | AAAS](#)) . This is a crux (Section 8): it is falsifiable in principle by additional early sequence data — several hundred early-patient sequences held by China remain unpublished ([Middle East Health](#)) .

3.3 The market’s animal evidence: “DNA and RNA ghosts”

The strongest market-specific forensic result came in September 2024, when Crits-Christoph, Débarre, Worobey and colleagues published in *Cell* a metagenomic reanalysis of environmental swabs collected in the market in early 2020. They found increased SARS-CoV-2 positivity near and within one wildlife stall, **wildlife DNA in all SARS-CoV-2-positive samples from that stall** — including raccoon dogs (the most abundant mammal in the wildlife-stall samples), masked palm civets, hoary bamboo rats and Malayan porcupines — plus animal viruses known to infect those species, and market-linked viral diversity “consistent with market emergence” ([Cell Press](#)) . Raccoon dogs are experimentally susceptible, shed high titers, and carried SARS-CoV-1 in 2003; civets were the amplifying host of the first SARS ([ScienceDaily](#)) .

The caveat is written into the paper’s provenance: “Many of the key animal species had been cleared out before the Chinese CDC teams arrived, so we can’t have direct proof that the animals were infected ... We are seeing the DNA and RNA ghosts of these animals in the environmental samples” (Débarre) ([ScienceDaily](#)) . Co-location of animal DNA and viral RNA on shared swabs establishes co-presence, not infection, and the market had been closed, cleared and disinfected before systematic sampling on 12 January and later — the sampling itself was of surfaces and drains, not animals ([ScienceDaily](#)) . China’s position, stated to the 2021 WHO mission, is that some **80,000 animals** across species and provinces were tested with no SARS-CoV-2 positives, and no animal sample from the market itself was positive ([Science | AAAS](#)) . The honest summary: the market was saturated with the virus and with susceptible animals in the same few stalls; whether the virus arrived in an animal, a trader, or a customer remains undetermined from these data alone.

3.4 The genome: relatives, the furin cleavage site, and the CGG-CGG argument

The closest known relatives of SARS-CoV-2 are bat coronaviruses: **RaTG13** (96.1–96.3% genome identity), sampled by the WIV in 2013 from a Tongguan mineshaft in Mojiang, Yunnan, and the **BANAL**

viruses from Laos (BANAL-52, 96.8%), identified in 2020 ([World Health Organization \(WHO\)](#)) . These are far too distant to be progenitors — RaTG13 differs from SARS-CoV-2 by roughly 1,100 nucleotides, implying decades of independent evolution — and none possesses the virus’s most discussed feature, the **furin cleavage site (FCS)**: a 12-nucleotide insertion (TCCTCGGCGGGC) encoding a PRRA polybasic motif at the spike S1/S2 junction that enhances cell entry and transmissibility ([Science | AAAS](#)) . SARS-CoV-2 is the only one of more than 800 known sarbecoviruses with an FCS, and the insertion’s two arginines are encoded by CGG-CGG, a codon doublet that accounts for only ~3% of arginine codons in the virus ([IntechOpen](#)) .

Each side of the FCS argument is well developed. For engineering-suspicion: the FCS is absent from every known close relative including the Laos/Vietnam viruses whose receptor-binding domains are otherwise near-identical; CGG-CGG is rare in coronaviruses but common in human codon usage and mammalian expression systems; Nobel laureate David Baltimore called it “the smoking gun for the origin of the virus” before walking the phrase back; and — critically — inserting human-specific cleavage sites into bat coronavirus spikes is precisely what the DEFUSE proposal described doing in Wuhan (Section 4.2) ([nih.gov](#)) . For natural acquisition: FCS motifs exist across other coronavirus lineages (HKU1, OC43, MERS-CoV and merbecoviruses), coronaviruses recombine promiscuously, the identical 12-nt sequence occurs in bat coronavirus Bat-HKU9, and the SARS-CoV-2 FCS is *suboptimal* — an engineer choosing from known sites would likely have used a more basic motif, per Susan Weiss and Stanley Perlman ([feedback.org](#)) . SAGO’s June 2025 report sided with natural recombination (“can be explained by natural recombination and mutations”), while a January 2026 peer-reviewed review concludes current evidence “does not permit definitive attribution” of the FCS ([World Health Organization \(WHO\)](#)) . Both can be true about the genome; they diverge about what the genome *implies*, and the genome alone will not settle it.

3.5 The two absences: no animal host, no lab progenitor

The zoonotic case carries one conspicuous hole: unlike SARS-CoV-1 (civets identified within months) and MERS (camels), no intermediate host carrying a SARS-CoV-2-like virus has been found after six years and tens of thousands of tested animals — a point SAGO itself flagged as “surprising, considering the many susceptible species present at [the market],” while cautioning that China’s animal testing has reported zero positives, which sits oddly beside the documented reverse-zoonosis of the virus into mink, deer and other species worldwide ([World Health Organization \(WHO\)](#)) . The absence is not dispositive — wildlife-farm upstream investigations were never completed and China has not shared animal-source data — but it is the principal empirical deficit of the market hypothesis ([WHO](#)) .

The lab case carries the mirror-image hole: no laboratory anywhere is known to have possessed SARS-CoV-2 or a sufficiently close progenitor before the outbreak; the ODNI reported in June 2023 “no indications that WIV research involved SARS-CoV-2 or a close progenitor virus,” and the intelligence community judges (low confidence) the virus was probably not genetically engineered ([FactCheck.org](#)) . That absence, too, is qualified by non-cooperation: the WIV’s own database of ~22,000 samples and sequences — including more than 100 unpublished bat-coronavirus sequences — went offline on 12 September 2019 and has never been shared, so “not found” and “not inspectable” are doing similar work on both sides ([The Washington Post](#)) . A rigorous seed must therefore treat the two absences symmetrically: each side’s strongest negative evidence is hostage to Chinese records that have never been produced.

3.6 Data-suppression episodes on the scientific record

Three episodes bear on the reliability of the *scientific* data itself, distinct from the political record. First, in June 2021 Jesse Bloom (Fred Hutchinson) recovered 13 early SARS-CoV-2 sequences that Wuhan University researchers had posted to the NIH Sequence Read Archive in March 2020 and then requested be deleted in June 2020; Bloom wrote that it “seems likely that the sequences were deleted to obscure their existence” ([Science | AAAS](#)) . The mundane explanation — the journal *Small* dropped the data-availability statement in copy-editing, and the authors withdrew the reads to avoid version-control problems before reposting to a Chinese database — was later corroborated by Wiley’s integrity team, and a 2024 re-examination concluded the samples were indeed collected on or after 30 January 2020, as the Chinese authors said ([bioRxiv](#)) . The recovered sequences nonetheless mattered scientifically: they lacked three mutations seen in market-linked viruses, supporting early circulation outside the market ([Science | AAAS](#)) .

Second, the WIV’s main coronavirus database — the most important bat-coronavirus database in China, holding records of ~22,000 samples, over 100 unpublished sequences, and the full sequences of eight viruses sampled in 2015 from the Mojiang region — became unreachable from outside the institute on 12 September 2019; Shi Zhengli says it was taken offline for security reasons after hacking attempts ([The Washington Post](#)) . Third, raw market environmental-sequence data were briefly posted to GISAID by Chinese CDC scientists in early 2023, enabling the Crits-Christoph analysis, then removed — the episode that prompted WHO to call China’s non-disclosure “inexcusable” ([Science | AAAS](#)) . Whatever the motive in each case, the pattern is that the open scientific record on both the animal and laboratory sides is incomplete specifically where Chinese custodianship is involved, a point SAGO formalized: China has not shared hundreds of early-patient sequences, detailed animal-source information, or laboratory biosafety documentation ([WHO](#)) .

4. The laboratory side: the Wuhan research program

4.1 The WIV–EcoHealth–NIH funding channel

The Wuhan Institute of Virology is China’s foremost SARS-coronavirus laboratory and sits roughly eight miles from the Huanan market; it held one of the world’s largest collections of bat-coronavirus samples ([New York Magazine](#)) . Between 2014 and 2019, the US NIH funded EcoHealth Alliance’s “Understanding the Risk of Bat Coronavirus Emergence” project, under which subawards flowed to the WIV — Fauci, in his June 2024 testimony, characterized the Wuhan subaward as “a \$120,000 a year grant in a \$6 billion budget” ([InfluenceWatch](#)) . In 2015, Ralph Baric (UNC) and Shi Zhengli (WIV) published NIAID-supported work combining a bat-coronavirus spike with a SARS backbone that could infect human airway cells — the experiment Rand Paul recited at the July 2026 hearing as “gained transmissibility. It gained function.” ([Rev](#))

Whether this work met the regulatory definition of gain-of-function became a years-long semantic war. Fauci’s consistent position, restated under oath in 2024, is that he used the “operative definition” under the HHS P3CO framework and that, on phylogenetics, “the viruses that were funded by the NIH ... could not be the precursor of SARS-CoV-2” — a narrow claim that is scientifically well supported but answers a different question than the one about unfunded or undisclosed work at the WIV, as Congressman Griffith’s exchange with Fauci (“maybe they used some of our money to get started, maybe they didn’t... a group of scientists getting together might very well at Wuhan have said, ‘Hey, let’s see what happens if we go over here and do this’”) made explicit, with Fauci conceding he “cannot account ... for other

things that might be going on in China” ([house.gov](#)) . The NIH’s own oversight record was later judged deficient by both parties: EcoHealth submitted a required progress report nearly two years late and failed to report an experiment in which modified bat coronaviruses replicated faster in humanized mice — findings the House cited and HHS later used as grounds for debarment ([Medpage Today](#)) .

4.2 DEFUSE: the 2018 proposal at the center of the circumstantial case

In March 2018, EcoHealth Alliance (Peter Daszak, PI), with UNC (Baric), the WIV (Shi Zhengli), Duke-NUS and others, submitted **Project DEFUSE** to DARPA’s PREEMPT program, requesting **\$14,209,245** for work in Kunming, Wuhan, Chapel Hill, New York, Singapore and elsewhere ([archive.org](#)) . The proposal described creating full-length infectious clones of bat SARS-related coronaviruses and inserting proteolytic cleavage sites — “of particular interest ... a type of cleavage site able to interact with furin” — into bat coronavirus spikes ([The Intercept](#)) . DARPA rejected it. When SARS-CoV-2 emerged twenty months later in the partner city, bearing exactly such a furin cleavage site, DEFUSE became the central documentary exhibit of the lab-leak case: as Paul put it at the July 2026 hearing, “this was either an extraordinary coincidence or a smoking gun” ([Rev](#)) .

The details that keep DEFUSE load-bearing are in the drafts. Margin comments released by U.S. Right to Know show Daszak telling Baric and Shi: “Once we get the funds, we can then allocate who does what exact work, and I believe that a lot of these assays can be done in Wuhan as well,” and an early draft specifying BSL-2 as “highly cost-effective” — to which Baric objected: “In the US, these recombinant SARS CoV are studied under BSL3, not BSL2 ... In china, might be growin these virus under bsl2. US researchers will likely freak out” ([Vanity Fair](#)) . At least fifteen federal agencies attended the DARPA PREEMPT Proposers Day where DEFUSE was pitched, yet the proposal’s existence was not public until 2021 ([senate.gov](#)) . And in April 2026 testimony released by Senator Paul in July, Baric told the Senate that the DEFUSE design “could have resulted in the creation of SARS-CoV-2,” that Shi could have performed the work alone without US involvement (“Research grants are always written in the context of being more fluid than contracts”), and that DARPA itself had solicited the cross-species-jumping research in January 2018 — though Baric also said he still believes an animal origin is “most likely” ([Washington Examiner](#)) . Newly obtained NIH records further indicate the DEFUSE concepts — spike swaps, receptor-binding changes, cleavage-site insertions — were embedded in multiple US-funded coronavirus projects years before the pandemic, making DEFUSE less an isolated bid than a visible specimen of an ongoing research program ([U.S. Right To Know](#)) .

4.3 Biosafety at the WIV

The declassified record on WIV biosafety is unusually specific. The ODNI’s June 2023 report states that **as of January 2019, WIV researchers performed SARS-like coronavirus experiments in BSL-2 laboratories**, “despite acknowledgements going back to 2017 of these viruses’ ability to directly infect humans through their spike protein and early 2019 warnings of the danger of this practice”; that some WIV researchers “probably did not use adequate biosafety precautions at least some of the time”; that a 2017 WIV dissertation demonstrated reverse-genetic cloning of SARS-like coronaviruses leaving **no traces of genetic modification**; that the WIV planned live-virus isolation from pangolin samples in fall 2019; and that a 2020 inspection found aging equipment, insufficient disinfectant capacity and ventilation problems ([Homeland Security Digital Library](#)) . Baric testified in January 2024 that he had personally warned Shi Zhengli to move the work to BSL-3 and she did not; he called the claim that WIV coronavirus work had adequate biosafety “a load of BS,” and recounted discovering that his humanized mice — shared under a use-restricted agreement — were being offered for sale by a Chinese commercial company ([Vanity Fair](#)) .

The counterweights also come from the same ODNI document: the IC knows of **no specific biosafety incident** that spurred the pandemic; the November 2019 WIV biosafety training course “appears routine, rather than an emergency response”; and China’s National Security Commission investigated the WIV in early 2020, drew blood from staff, and — per WIV officials including Shi — found no SARS-CoV-2 antibodies ([Homeland Security Digital Library](#)) . The biosafety evidence therefore establishes *opportunity and degraded practice*, not *an event*. That is why it functions as a circumstantial multiplier for the lab hypothesis rather than a proof, and why the lab case keeps returning to the sealed records: staff health files, raw sequence logs, and experiment notebooks are exactly the documents that would convert opportunity into either an incident or an exoneration ([CBS News](#)) .

4.4 Sick researchers and the offline database, autumn 2019

The State Department’s January 2021 fact sheet asserted that the US government “has reason to believe that several researchers inside the WIV became sick in autumn 2019, before the first identified case of the outbreak, with symptoms consistent with both COVID-19 and common seasonal illnesses” ([The Washington Post](#)) . The ODNI’s June 2023 report deflated the sharpest version of this: several WIV researchers were indeed ill in fall 2019, but with “a range of symptoms consistent with colds or allergies,” some confirmed to have had unrelated illnesses, none known to have been hospitalized for COVID-like symptoms (one may have been hospitalized for a non-respiratory condition), and the IC “is unable to confirm if any of them handled live viruses” — concluding the episode “neither supports nor refutes either hypothesis” ([Homeland Security Digital Library](#)) . The September 2019 database takedown is firmer as a fact and softer as an inference: it happened on 12 September 2019, three months before the outbreak, it removed ~22,000 sample records from outside view, and the offered explanation — hacking — has never been verifiable ([The Washington Post](#)) . Neither item is proof; together with the biosafety record they form the “opportunity” leg of the circumstantial case.

4.5 The Mojiang miners and RaTG13

The deepest backstory link between the WIV and severe pneumonia predates the pandemic by seven years. In April 2012, six miners clearing bat feces from the Tongguan mineshaft in Mojiang, Yunnan, developed severe atypical pneumonia; **three died**; a Kunming medical master’s thesis concluded a SARS-like coronavirus from horseshoe bats was the likely agent, and prominent pulmonologist Zhong Nanshan consulted remotely ([Frontiers](#)) . The WIV sampled the mine repeatedly between 2012 and 2015, isolating 293 coronaviruses — among them, in 2013, the sample later renamed **RaTG13**, the closest known relative of SARS-CoV-2 ([wikipedia.org](#)) . Shi Zhengli attributed the miners’ deaths to fungus in a 2020 *Scientific American* interview, said the miners’ serum (sent to WIV) tested negative for SARS-CoV antibodies, and did not disclose the mine–RaTG13 link in print until July 2020 ([Frontiers](#)) .

What Mojiang proves is contested in both directions. For the lab side, it shows the WIV surveilling a site where a SARS-like virus plausibly killed humans, holding unpublished sequences from it, and being less than forthright about the connection ([Frontiers](#)) . For the zoonotic side, the miners’ illness is equally a demonstration that bat-coronavirus spillover into humans *happens in nature without any laboratory* — the ecological prior on which the market case rests — and WIV retesting of miners’ serum in 2020 for SARS-CoV-2 was negative ([wikipedia.org](#)) . The mine’s eight unpublished 2015 virus sequences sit in the offline database; they are one of the concrete data items whose release would bear directly on the progenitor question ([The Washington Post](#)) .

5. The hearings and the circumstantial record, 2023–2026

This section is the record the user asked to be covered in depth: what the recent congressional hearings, document releases, and intelligence actions actually established — distinguishing (a) evidence about the virus from (b) evidence about the conduct of officials and scientists, which is where the new material almost entirely falls.

5.1 The 1 February 2020 teleconference and the making of “Proximal Origin”

On 31 January 2020, Kristian Andersen emailed Fauci that “the unusual features of the virus make up a really small part of the genome (<0.1%) so one has to look really closely at all the sequences to see that some of the features (potentially) look engineered,” adding that he, Eddie Holmes, Bob Garry and Mike Farzan “all find the genome inconsistent with expectations from evolutionary theory” ([USA Today](#)) . The next day, Jeremy Farrar convened a confidential teleconference — Fauci, Francis Collins, Andersen, Holmes, Garry, Andrew Rambaut, Ron Fouchier, Patrick Vallance and others — which participants agreed to treat “in total confidence”; that morning Fauci emailed his deputy Hugh Auchincloss: “It is essential that we speak this AM ... You will have tasks today that must be done,” later explaining he wanted to know “the scope of what our collaborations were and the kind of work that we were funding in China” ([The Nation](#)) . On the call, per later testimony, Farzan put lab origin at 70/30, Farrar at 50-50, and Holmes at 60-40 leaning lab ([U.S. Right To Know](#)) .

Within four days the group’s public posture inverted. On 4 February, Andersen told the National Academies drafting group that “the data conclusively show” no engineering, calling such theories “crackpot” — while specifying to Congress later that he meant engineered *with intent*, and that he personally kept a lab-culture possibility open until mid-to-late February ([USA Today](#)) . On 4 February Farrar sent Fauci and Collins a “very rough first draft from Eddie and team”; on 8 February Andersen wrote that “our main work over the past couple of weeks has been to **disprove** any type of lab theory, but we are at a crossroad where the scientific evidence isn’t conclusive enough”; on 12 February he pitched the paper to *Nature* as rebutting “speculation, fear mongering, and conspiracies,” invoking Farrar, Fauci and Collins by name; and on 17 February, minutes before the preprint posted, Farrar requested the key sentence be softened from “unlikely” to “improbable” that SARS-CoV-2 emerged through laboratory manipulation ([U.S. Right To Know](#)) . The paper — “The Proximal Origin of SARS-CoV-2,” *Nature Medicine*, 17 March 2020 — concluded “we do not believe that any type of laboratory-based scenario is plausible,” has been accessed more than 5.6 million times, and became the load-bearing citation for dismissing the lab hypothesis ([The Intercept](#)) .

5.2 The Lancet letter and the suppression claim

On 19 February 2020, *The Lancet* published a letter from 27 scientists “strongly condemn[ing] conspiracy theories suggesting that COVID-19 does not have a natural origin,” with all signatories declaring no competing interests ([wikipedia.org](#)) . FOIA emails later showed the letter was organized by Daszak, who deliberately omitted both EcoHealth’s WIV partnership and Baric’s name to feign impartiality; *The Lancet* added a competing-interests addendum in June 2021 after sustained criticism, and editor Richard Horton conceded Daszak “should certainly have declared his competing interests right at the beginning” ([LibGuides at Memorial Sloan Kettering Cancer Center](#)) . The letter’s effect on discourse was substantial: Vanity Fair’s Katherine Eban described its “chilling effect,” and critics including Nicholas Wade noted the signatories were “assuring the public of facts they could not know for sure were true” ([wikipedia.org](#)) .

The suppression claim is one place where the record must be kept honest in both directions. It is

documented that Proximal Origin’s authors publicly expressed more confidence than they privately held (Section 5.8), that the Lancet letter concealed its organizer’s conflict, and that officials discussed “putting down” the lab-leak “conspiracy”: on 16 April 2020 Collins emailed Fauci asking what more NIH could do “to help put down this very destructive conspiracy,” and the next day Fauci cited Proximal Origin from the White House podium as showing the virus “is totally consistent with a jump of a species from an animal to a human,” without disclosing his role in prompting the paper (Rev) . It is equally documented that Fauci’s own contemporaneous email to Farrar on 1 February 2020 directed that if the concerns were validated they should be reported “to the appropriate authorities ... in the U.S.A., this would be the FBI” — the opposite of burying the concern — and that the scientists’ shift was accompanied by genuine new analyses (proximity of the outbreak to the market, RBD recombination analyses), which authors describe as the scientific process working (Rev) . Both halves belong in the seed; the crux is not whether views changed but whether the *public certainty* was ever warranted by the *private evidence*.

5.3 Morens and the FOIA-evasion record (May 2024)

The May 2024 hearing of Dr. David Morens — Fauci’s senior scientific adviser for decades — produced the investigation’s most damaging primary documents. Morens wrote on 24 February 2021: “i learned from our foia lady here how to make emails disappear after i am foia’d but before the search starts, so i think we are all safe. Plus i deleted most of those earlier emails after sending them to gmail” (kffhealthnews.org) . Under questioning he admitted using personal email for official business to avoid FOIA (“I shouldn’t have done that. That’s wrong”), was confronted with evidence of intentional misspellings to defeat keyword searches and of backchannel routes to Fauci, and was told by Rep. McCormick that, given conflicting statements, he had lied either in his transcribed interview or at the hearing (house.gov) . Both parties condemned the conduct; Chairman Wenstrup wrote to NIH that the evidence “suggests a conspiracy at the highest levels of NIH and NIAID to avoid public transparency,” while Democrats stressed the documents implicate records practices, not the virus (kffhealthnews.org) .

Morens’s emails also show him backchanneling confidential NIH grant information to Daszak — his “best-friend” — while EcoHealth’s grant was under review (house.gov) . The hearing’s significance for the origins case is indirect but real: it demonstrated that senior NIAID staff handled origins-related correspondence through channels designed to be invisible to FOIA and to Congress, which means the public documentary record of early-2020 deliberations was, by design, incomplete. Congressman Griffith drew the civil-law analogy explicitly at the June 2024 Fauci hearing: “when one party has destroyed or refuses to produce evidence ... a jury is allowed to draw an adverse inference that the information destroyed or not produced was unfavorable” (Rev) . Whether that inference is fair is contested; that the deletions occurred is not (house.gov) .

5.4 Fauci’s transcribed interview (January 2024) and public hearing (June 2024)

Across a two-day, 14-hour transcribed interview in January 2024, Fauci made several concessions the subcommittee later weaponized and Democrats later contextualized: the six-foot distancing rule “sort of just appeared” without supporting trials; vaccine mandates may have increased hesitancy; he said “I do not recall” more than 100 times; and — the headline — **the lab-leak hypothesis “in and of itself isn’t inherently a conspiracy theory,”** a marked retreat from the 2020–2021 discourse his own circle had shaped (house.gov) . On gain-of-function he insisted on the P3CO “operative definition,” the semantic position that underlay his 2021 Senate denials that NIH funded gain-of-function research in Wuhan (house.gov) .

At the 3 June 2024 public hearing, Fauci denied editing or prompting the content of Proximal Origin

(“The accusation ... that I influenced these scientists to change their minds by bribing them with millions of dollars in grant money is absolutely false and simply preposterous”), denied deleting records, and held his two-part position: NIH-funded viruses “phylogenetically could not be the precursor of SARS-CoV-2,” while “I cannot account nor can anyone account for other things that might be going on in China ... I keep an open mind” ([Rev](#)) . Republicans pressed the adverse-inference argument from the deleted-records record; Democrats entered a minority report (“Republicans’ Fauci Flop”) concluding the probe “fails to find evidence” for the extreme claims; and the hearing transcript preserves the exchange in which Fauci both affirmed Proximal Origin’s authority and conceded a passage-origin scenario “could be” possible — the same paper having dismissed “any type of laboratory-based scenario” ([Rev](#)) .

5.5 The Select Subcommittee final report (December 2024) and the minority response

On 2 December 2024, the Republican majority released its **520-page final report** after 25 hearings, 30+ transcribed interviews and one million pages of documents, concluding COVID-19 “most likely emerged from a laboratory in Wuhan” on five argued grounds: the FCS “not found in nature”; all cases from “a single introduction”; Wuhan’s SARS lab working at “inadequate biosafety levels”; sick WIV researchers in fall 2019; and the non-appearance of natural-origin evidence ([house.gov](#)) . The report also concluded Fauci “prompted” Proximal Origin “to push the preferred narrative,” that EcoHealth “facilitated dangerous gain-of-function research ... and misled the NIH,” and — newly revealed — that the **Department of Justice had empaneled a grand jury** connected to COVID-19 origins, subpoenaing EcoHealth documents, an investigation whose details and status remain unknown ([The Hill](#)) .

Two cautions are due on this report. It contains **no new direct evidence** of a lab leak — Science’s characterization — and its five signature claims are contested by the scientific record: the FCS does exist in other coronavirus lineages, and the two-lineage analysis argues against “a single introduction” ([Science](#) | [AAAS](#)) . The Democratic minority report concluded “a zoonotic origin and lab accident are both plausible, as is a ‘hybrid’ scenario,” defended Fauci, and noted that WIV viruses studied with EcoHealth funding were too distantly related to be the progenitor ([The BMJ](#)) . Fact-checkers’ verdict on the five-point formulation (later reused verbatim by the White House webpage, Section 6.2): each point is “incorrect or misleading” in its strong form, even though the underlying circumstantial questions are legitimate ([FactCheck.org](#)) .

5.6 Debarment, the pardon, and legal jeopardy

On 17 January 2025 — the Biden administration’s final days — HHS **formally debarred EcoHealth Alliance and Peter Daszak for five years**, citing the subcommittee-uncovered record: ignored oversight requests, failure to report the humanized-mouse experiment, and a progress report two years late; EcoHealth had terminated Daszak effective 6 January 2025 ([house.gov](#)) . Senator Kennedy characterized the cited experiments as modifying bat coronaviruses “to make them 10,000 times more infectious in mice” ([U.S. Senator John Kennedy](#)) . On 20 January 2025, President Biden issued Fauci a **preemptive pardon** covering any federal offenses from 1 January 2014 through 19 January 2025 — a shield with holes, as later legal analysis noted: it does not cover perjury after 19 January 2025 or state offenses, and three state attorneys general subsequently subpoenaed Fauci ([The Hill](#)) .

The legal posture matters to the case only as context for 2026: it is why, when Rand Paul subpoenaed him, Fauci’s lawyers judged that any sworn answer could seed a perjury theory despite the pardon, and why the contempt fight turned on whether a pardoned witness retains Fifth Amendment protection — with former prosecutors split, and Democrats warning the referral “will create a damaging precedent” for oversight generally ([The Hill](#)) . Whatever one’s view of the merits, the sequence — debarment of the

grantee, pardon of the official, grand-jury empanelment of unknown scope — means the US legal system, not just its political system, now has live threads into the origins question ([The Hill](#)) .

5.7 The June 2026 ODNI “Fauci Files”

On 18–19 June 2026, in one of her final acts as Director of National Intelligence, Tulsi Gabbard released **67 documents across 391 pages**, headlined “Fauci Funded Wuhan Lab Research That Sparked COVID,” alleging Fauci lied to Congress under oath in 2024, manipulated intelligence assessments, and ran a “deep state playbook” ([WBMA](#)) . The genuine news value in the drop is that it documents Fauci’s participation in a February 2020 expert meeting on data needs and a June 2021 IC briefing in which he discussed sick WIV researchers and recommended scientists to consult ([lawfaremedia.org](#)) .

Lawfare’s document-level review is the essential counterweight and should be cited alongside the release itself: the files contain no intelligence assessment concluding Fauci-funded research “sparked COVID”; the CIA’s internal email states it has **“no record of Dr. Fauci coming to CIA headquarters”** to discuss origins (contradicting Paul’s implication); IC officials considered Fauci as an outside reviewer and **rejected him as conflicted**; after a contentious House briefing an ODNI official wrote it was “laughable” to claim lab-leaning experts were excluded — “every expert they named was consulted”; and the “analysts were paid off” whistleblower claim is paired in the file with its alternative explanation (late-arriving information) ([lawfaremedia.org](#)) . Independent fact-checkers reached the same bottom line: across ~400 pages there is no definitive link proving Fauci knowingly concealed an origin, and no new evidence for either hypothesis ([WBMA](#)) . The drop is thus strongest evidence for a *meta* claim — that origins intelligence work and political actors were entangled — and weak evidence for any *origin* claim.

5.8 The Proximal Origin Slack messages (22 July 2026)

On 22 July 2026, Paul released internal Slack messages among the Proximal Origin authors showing that, three months after publishing that no laboratory scenario was plausible, **Andersen assigned a 30% probability to a lab origin, Holmes 20% (later revised to 10%)**; Andersen wrote “our paper was pretty strong in saying ‘there’s no way,’ but I have less confidence in that statement at this stage,” and reacted to German researchers building a SARS-CoV-2 reverse-genetics system in under two weeks with “Clearly it’s not that hard” ([senate.gov](#)) . The messages also show the group coordinating with the intelligence community during its 90-day review (Andersen expecting to “have a handler”), Holmes repeatedly citing a Chinese source he called “my spy,” and — after DEFUSE leaked — discussing the need to “stay off email,” to “carefully curate” correspondence “for the FOI records,” and to “get something back to Tony [Fauci]’s team at NIH” ([senate.gov](#)) .

Interpretation splits predictably and both readings should be preserved. To investigators, this is the suppression claim confirmed in the authors’ own words: public certainty outrunning private belief, records curated against FOIA, and an NIH feedback loop ([senate.gov](#)) . To the authors’ defenders, private probability estimates of 10–30% are what scientific uncertainty looks like; the paper’s conclusion concerned deliberate engineering (“not a laboratory construct or a purposefully manipulated virus”), which the Slack messages did not resurrect; and holding a minority-probability scenario while publishing the best-supported one is normal science, not concealment ([The Conversation](#)) . What the messages indisputably change is the *status of the paper as a consensus marker*: the document most often cited as “the science has spoken” was written by people who, in their own contemporaneous words, had not finished speaking ([senate.gov](#)) .

5.9 The Fauci diary, the Fifth Amendment hearing (29 July 2026), and contempt (6 August 2026)

Days before the hearing, Paul released **over 1,000 pages of Fauci’s private diary entries (December 2019–2022)** plus a 465-page “prequel” of records from 2001–2015 (NPR) . The entry Paul foregrounded, from 1 February 2020, records of the Farrar call that “there was not total agreement” about origins; elsewhere Fauci wrote he “is almost certain that the virus evolved naturally from a species jump, even though ... I keep an open mind about the possibility of a lab leak,” and “that just because two alternatives are possible that does not mean that they are equally probable” (NPR) . Read one way, the diary shows private uncertainty behind public assurance; read the other, it shows exactly the calibrated view Fauci later testified to. Virologists Felicia Goodrum and Gigi Gronvall made the latter case to NPR: early-2020 entertaining of lab scenarios was “normal scientific questioning,” now retro-presented as “some sort of signal of guilt” (NPR) .

At the 29 July 2026 Senate HSGAC hearing, appearing under subpoena and for the first time since the pardon, Fauci **invoked the Fifth Amendment to every question — 111 times**, including “What color tie are you wearing?”; his lawyer was ejected after attempting to interject; Paul announced a contempt vote, arguing the pardon voids the privilege (PBS) . On 6 August 2026 the committee voted **8–5 (with two more opposed by proxy) to hold Fauci in contempt**, and Paul sent the referral directly to the Justice Department, bypassing a floor vote; the resolution directs that the report be certified to US Attorney Jeanine Pirro, with DOJ to decide on charges (CBS News) . Former prosecutors described the case as “incredibly messy”: Fauci “would have a bulletproof motion to dismiss on Fifth Amendment grounds” given Paul’s stated aim of seeing him “behind bars,” while the pardon’s gaps (post-January-2025 perjury, state crimes, three pending state-AG subpoenas) arguably preserve a “real and appreciable” risk that sustains the privilege (The Hill) . For the case file: the hearing produced **no new origin evidence**; it is evidence about accountability and about how contested the documentary record has become.

6. Governments and intelligence services

6.1 The assessment table

The institutional landscape as of August 2026, with each body’s stated confidence language, is consolidated below. Two structural observations belong in the seed: every lab-leaning assessment is explicitly **low or moderate confidence** (except BND’s reported 80–95%), and the two most scientifically granular assessments (SAGO; the *Science/Cell* literature) lean natural while depending almost entirely on Chinese-supplied data that China has partially withheld (NBC News) .

Body	Position	Confidence	Date	Basis (as stated)
Mainstream virology literature (<i>Science</i> 2022; <i>Cell</i> 2024)	Market zoonosis	High (authors)	2022–2024	Case clustering; two lineages; wildlife DNA co-location [(Science
WHO SAGO (27 experts)	Zoonotic spillover suggested; lab not excludable, not assessable	“Weight of available evidence”	Jun 2025	Review of literature + government/ intelligence reports; China data gaps (WHO)

Table 2 – continued

Body	Position	Confidence	Date	Basis (as stated)
US NIC + 4 IC entities	Natural exposure	Low	Oct 2021	2021 declassified assessment (FactCheck.org)
FBI	Lab incident, probably WIV experimentation/handling/sampling	Moderate	2021; public 2023	Wray: “most likely a potential lab incident in Wuhan” (ABC News)
US Dept. of Energy	Lab leak more likely	Low	Feb 2023	National-laboratory analysis, reportedly new intelligence (NPR)
CIA	Research-related origin more likely	Low	Jan 2025	Re-review of existing reporting; ordered under Burns, declassified under Ratcliffe (NBC News)
BND (Germany)	Lab accident	80–95% probability (reported)	2020; revealed Mar 2025	“Saaremaa” operation data incl. Chinese research material; not published by Merkel govt. (The Telegraph)
ODNI (community-wide)	No determination; not a bioweapon; probably not engineered (low conf.); no pre-outbreak Chinese knowledge	—	Oct 2021 / Jun 2023	Declassified reports (FactCheck.org)
House Select Subcmte (GOP)	Lab origin “most likely”	Political judgment	Dec 2024	520-page report; five circumstantial arguments (house.gov)
House Select Subcmte (Dem.)	Both plausible; hybrid possible	Political judgment	Dec 2024	Minority report (The BMJ)
China (NHC / State Council)	Lab leak “extremely unlikely”; natural origin or cold chain; hints at US origin	Official position	2021; Apr 2025	WHO-China joint study; white paper (国家卫生健康委员会)
US White House (Trump)	Lab leak “the true origins”	Official position	Apr 2025	Webpage adopting subcommittee’s five points (NPR)

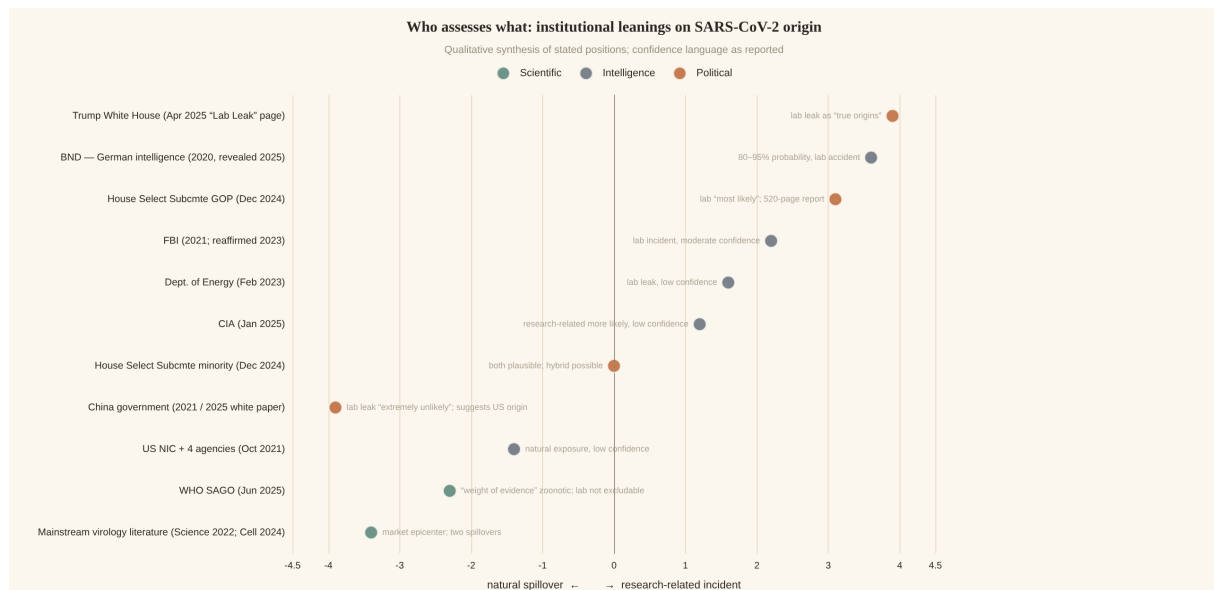


Figure 2: Institutional leanings

Figure 2. Qualitative placement of institutional positions, colored by type. The vertical dashed line is “undecided.” Note the band structure: science leans left on direct data, intelligence leans right on classified and circumstantial reporting, political bodies sit at the poles.

6.2 The politicization layer (2025–2026)

On 18 April 2025 the White House redirected covid.gov to a page titled “**Lab Leak: The True Origins of COVID-19,**” presenting the subcommittee’s five points and accusing Fauci of pushing a “preferred narrative”; virologist Angela Rasmussen responded that “every one of the five pieces of evidence ... is factually incorrect, embellished, or presented in a misleading way,” and FactCheck.org documented that the administration’s claim of a “confirmed” lab origin misstates the agencies’ low/moderate-confidence language ([ABC News](#)) . China answered on 30 April 2025 with a State Council white paper claiming the virus may have circulated in the US earlier — citing July 2019 Virginia pneumonia reports, Florida data, 28 UNC lab incidents, and the 2019 Fort Detrick shutdown — and declaring China’s own investigation complete ([BBC Monitoring](#)) . A Missouri federal court’s 7 March 2025 default judgment ordering China to pay **\$24.49 billion** added a judicial-political thread Beijing rejects outright ([The State Council Information Office of China](#)) .

For the case file, the politicization layer matters because it changed what “official position” means: since January 2025, the US government’s *political* organs assert lab origin as settled while its *intelligence* organs hold it at low-to-moderate confidence and its *scientific* literature leans the other way. Any honest claim ladder must therefore keep these three evidentiary systems separate rather than averaging them ([NBC News](#)) .

7. Claim ladder with evidence mapping

The ladder below is the seed’s core: each atomic claim, the best evidence for and against, and a status. “Conduct” claims (C012–C016) are included because the hearings made them load-bearing for public trust in the origin claims themselves, though they do not locate the virus.

ID	Atomic claim	For	Against / caveat	Status
COV-C001	Huanan market was the early epicenter	66% of first 41 cases exposed; 55/168 linked; unlinked cases lived closer than linked; excess pneumonia deaths rose first nearby; positive samples cluster in wildlife section (The Lancet)	Earliest case (Dec 1) had no market link; market may be amplifier not origin (SAGO agnostic) (The Lancet)	Supported as epicenter; Contested as origin
COV-C002	Two lineages two spillovers	Phylogenetic simulations reject single-introduction pattern; B then A within days-weeks [(Science	AAAS)](https://www.science.org/doi/10.1126/science.abp8337)	Rooting-sensitive; early sampling sparse and partly withheld; a single jump with early diversification not fully excludable [(Science
COV-C003	Susceptible wildlife + virus co-located	Wildlife DNA in all virus-positive samples from one stall; raccoon dogs abundant; animal viruses present (Cell Press)	Animals cleared before sampling; co-presence infection; zero animal samples positive (ScienceDaily)	Supported (presence only)
COV-C004	An infected intermediate host exists	SARS-1/MERS precedent; susceptible species documented at market (ScienceDaily)	~80,000 animals tested, none positive; six years, no host found; upstream farm investigation never done [(Science	AAAS)](https://www.science.org/content/article/it-s-inexcusable-who-blasts-china-not-disclosing-potential-data-covid-19-s-origin)
COV-C005	No lab held a progenitor	ODNI: no indications WIV had SARS-CoV-2 or close progenitor; IC: probably not engineered (Fortune)	WIV database (22k samples, 100+ unpublished sequences) offline since 12 Sep 2019; staff health records sealed; seamless-cloning capability documented (The Washington Post)	Unresolved (unverifiable)

Table 3 – continued

ID	Atomic claim	For	Against / caveat	Status
COV-C006	FCS is anomalous/ engineered-looking	Unique among 800+ sarbecoviruses; CGG-CGG rare (3%); Baltimore “smoking gun” remark; matches DEFUSE plan (IntechOpen)	FCS in other CoV lineages; identical 12-nt in Bat-HKU9; site is suboptimal for engineering; natural recombination plausible (SAGO) (feedback.org)	Fact Supported ; inference Contested
COV-C007	DEFUSE shows intent/capability to make SARS-CoV-2-like virus	2018 proposal: FCS insertion, infectious clones, WIV partner; BSL-2 draft language; Baric 2026: could have created such a virus; Shi could have done it alone (archive.org)	Never funded by DARPA; no evidence the work occurred; Baric himself leans animal origin (The Intercept)	Fact Supported ; relevance Contested
COV-C008	WIV did SARS-like CoV work at BSL-2	ODNI 2023: BSL-2 work as of Jan 2019 despite 2017 human-infectivity findings; Baric warnings ignored; “load of BS” email (Homeland Security Digital Library)	No known specific incident; Nov 2019 training “routine”; staff serology negative per WIV (Homeland Security Digital Library)	Supported (practice), no incident shown
COV-C009	Database takedown was suspicious	Offline 12 Sep 2019, three months pre-outbreak; never restored to outsiders; includes Mojiang-related sequences (The Washington Post)	Hacking explanation offered; timing could coincide with routine security review (BuzzFeed News)	Fact Supported ; motive Contested
COV-C010	WIV staff illness was early COVID	State Dept. fact sheet: several sick, autumn 2019, COVID-consistent symptoms (The Washington Post)	ODNI: symptoms matched colds/ allergies, some confirmed other illnesses, no COVID hospitalizations; “neither supports nor refutes” (Homeland Security Digital Library)	Unresolved

Table 3 – continued

ID	Atomic claim	For	Against / caveat	Status
COV-C011	No circulation before ~Nov 2019	Two-lineage bound; no verified pre-Dec human/animal positives (SAGO) [(Science	AAAS)](https://www.science.org/doi/10.1126/science.abp8337)	Early Chinese sequences unpublished (500+); deleted-sequence episode showed record fragility (Middle East Health)
COV-C012	Public certainty exceeded private belief	Andersen 30%/Holmes 20%→10%; “less confidence in that statement”; diary: “not total agreement”; FOIA-curation chat (senate.gov)	Paper targeted deliberate engineering; uncertainty is normal science; views updated on new data (National Review)	Supported (documentary)
COV-C013	NIAID figures evaded records laws	Morens “make emails disappear” email; deleted records; FOIA-lady instruction; admitted on record (kffhealthnews.org)	Conduct origin; no evidence Fauci personally deleted records (denied under oath) (Rev)	Supported (Morens), Contradicted as to Fauci personally
COV-C014	Fauci “prompted” Proximal Origin to kill the lab theory	Subcommittee finding; Collins “put down this very destructive conspiracy”; podium citation next day (house.gov)	Fauci’ s Feb 1 email urged reporting concerns to FBI; authors deny direction; Feb 2020 diary entry shows open consideration (Rev)	Contested
COV-C015	NIH funded gain-of-function research in Wuhan	2015 chimeric spike work; humanized-mouse experiment unreported; HHS debarment rationale (Rev)	P3CO “operative definition” excludes it per NIH/Fauci; funded viruses phylogenetically unrelated to SARS-CoV-2 (undisputed) (house.gov)	Contested (definitional)

Table 3 – continued

ID	Atomic claim	For	Against / caveat	Status
COV-C016	Intelligence manipulation by health officials	Gabbard release: Fauci briefed IC, recommended experts (lawfaremedia.org)	CIA: no record of HQ visit; IC rejected Fauci as conflicted reviewer; “every expert they named was consulted” (lawfaremedia.org)	Contradicted (on current documents)



Figure 3: Evidence balance across the claim ladder

Figure 3. Diverging qualitative weights across the seed claims. Grey bars (C010, C012) mark items that are ambiguous or bear on conduct rather than the virus. Weights are the seed author’s synthesis of the cited record, intended as starting values for the atlas’s own scoring — not measurements.

8. Where the disagreement lives

Crux 1 — epicenter vs. origin. Both sides now accept the market was the early epicenter; the dispute is whether the market was the *spillover site* (zoonosis) or a *superspreading node* seeded by an infected person (lab incident). The evidence distinguishing these is thin because the market was cleared and disinfected before animal sampling, and the animal-source farms were never investigated ([Science | AAAS](#)). This crux is resolvable in principle: wildlife-farm traceback data and the animals’ supply-chain records exist in China and have never been shared ([Middle East Health](#)).

Crux 2 — one introduction or two. The two-lineage result currently does the most work for zoonosis (SAGO treats it as a signature “rather than ... a single virus introduction”), and it is also the result most sensitive to data withheld from it — rooting, missing early genomes, the 500+ unpublished sequences ([World Health Organization \(WHO\)](#)). If released early genomes collapse A and B into one introduction,

the zoonotic case loses its strongest quantitative pillar; if they reinforce two market-linked introductions, the lab case loses its “single introduction” argument.

Crux 3 — absence of evidence on both sides. No infected intermediate host after ~80,000 animals; no progenitor in any declared collection. Each side reads the other’s absence as telling and its own as merely hard ([Science](#) | [AAAS](#)) . The asymmetry that can break the tie is custodianship: the lab side’s absence is potentially remediable by WIV records (database, notebooks, health files); the animal side’s absence is remediable by farm/trade records. Both sets are in Chinese hands, which is why SAGO’s report reads simultaneously as a zoonosis-leaning assessment and an indictment of the data environment ([WHO](#)) .

Crux 4 — the FCS as signal or noise. The genome cannot adjudicate itself: identical features (unique-among-sarbecoviruses FCS, rare CGG-CGG, suboptimal motif) are read as anomalous by one side and as within natural variation by the other, with serious scientists on both sides (Baltimore’s “smoking gun” remark and partial walkback; Weiss/Perlman’s suboptimality argument) ([California Institute of Technology](#)) . This crux would yield to phylogenetic discovery — a natural sarbecovirus with an FCS — or to documentary discovery of FCS-insertion work at WIV, not to further argument about the sequence alone ([IntechOpen](#)) .

Crux 5 — what the conduct record proves. The hearings documented real misconduct (records evasion, conflict-concealed letters, public-private divergence) ([house.gov](#)) . The unresolved interpretive question is whether that conduct was (a) evidence-conscious concealment of a feared lab origin, or (b) ordinary institutional self-protection under unprecedented scrutiny — and, critically, whether either answer changes the *virological* evidence. The seed should keep this crux explicitly orthogonal: misconduct can justify re-weighting testimony without constituting origin evidence.

9. What would settle it — the research agenda

Ordered by decisiveness-per-feasibility, with provenance demands noted. This section is written for Aletheia’s “Price of Truth” layer: each test names the evidence, the custodian, and the claim IDs it moves.

#	Decisive test / dataset	Custodian	Moves claims	Expected yield
T1	Release of WIV’ s offline database (22,000 samples; 100+ unpublished sequences; 8 Mojiang-related genomes)	WIV / CAS	C005, C006, C009	High. Either a near-progenitor appears (case-transforming) or the progenitor search space narrows materially (The Washington Post)
T2	Release of 500+ unpublished early-patient sequences and raw Dec-2019 case data	China CDC / NHC	C001, C002, C011	High. Directly tests one-vs-two introductions and pre-December circulation (Middle East Health)

Table 4 – continued

#	Decisive test / dataset	Custodian	Moves claims	Expected yield
T3	Wildlife supply-chain investigation: farms and traders supplying Huanan' s western section, with serology of 2019 stock and workers	China (agriculture authorities)	C003, C004	High if animals infected; else inconclusive. SAGO' s named priority (WHO)
T4	Independent audit of WIV experiment logs, sample inventories, and staff health records (2018–2019), incl. serology archive drawn by China' s NSC in early 2020	WIV / Chinese govt.	C005, C008, C010	High. The only path to confirming/ excluding a laboratory-acquired infection (Homeland Security Digital Library)
T5	Broadened sarbecovirus sampling in bats/ intermediate hosts across SE Asia targeting FCS-bearing relatives	International field programs	C006	Medium-high over years. A natural FCS-bearing sarbecovirus would neutralize the “unique feature” argument (IntechOpen)
T6	Full declassification of FBI/DOE/CIA/ BND underlying reporting (not just leanings)	US DOJ/DOE/CIA; German govt.	All L-family claims	Medium. Reveals whether classified case rests on signals/ human intelligence or re-weighted open science (NBC News)
T7	Release of DOJ grand-jury findings re: EcoHealth (if any)	US DOJ	C013–C015	Medium. Legal record on grant misconduct; tangential to origin but probative on conduct (The Hill)

Table 4 – continued

#	Decisive test / dataset	Custodian	Moves claims	Expected yield
T8	Adversarial, preregistered re-analysis of the two-lineage question with all available early genomes (proponents and critics fixing methods before unblinding)	Scientific community	C002	Medium. Same design Aletheia applied to CCC; removes method-shopping suspicion [(Science

A note on expected cost: T1–T4 and T6–T7 are acts of political will, not research programs — their cost is diplomatic, and their current probability is low given that China has declared origins-tracing “finished” and the US has replaced scientific outreach with an accusatory webpage ([Bulletin of the Atomic Scientists](#)) . T5 and T8 are genuine research items, fundable at field-survey and analysis scale respectively. The single highest-value item remains **T1**: the database whose takedown three months before the outbreak is the case’s most concrete unexplained act ([The Washington Post](#)) .

10. Load-bearing claims and weakest links

Load-bearing for zoonosis: COV-C001 (epicenter clustering — the sturdiest item in the whole case), COV-C002 (two introductions), COV-C003 (wildlife co-location). If C002 falls to released data, the zoonotic case retreats to “market amplification + unknown upstream event,” materially weakening it against the lab scenario ([Science | AAAS](#)) .

Load-bearing for a research-related origin: COV-C007 (DEFUSE’s existence and content), COV-C008 (BSL-2 practice), COV-C005 (unverifiability of lab holdings). The lab case currently has no direct positive evidence — no progenitor, no incident, no whistleblower; its strength is the conjunction of opportunity, capability, intent-on-paper, and opacity ([archive.org](#)) .

Weakest links on each side: For zoonosis — COV-C004 (the missing host; six years, ~80,000 animals, zero positives) and the dependence of market metagenomics on post-clearance surface swabs ([Science | AAAS](#)) . For the lab case — the sick-researchers item (COV-C010), which the ODNI explicitly says “neither supports nor refutes either hypothesis,” and the FCS-uniqueness argument in its strong “not found in nature” form, which is factually wrong as stated on the White House page ([Homeland Security Digital Library](#)) . For the meta-claims — the suppression narrative’s strong version (COV-C016, intelligence manipulation) is currently **contradicted** by the very documents released to support it, while its moderate version (COV-C012, public certainty exceeding private belief) is **supported** ([lawfaremedia.org](#)) .

Suggested initial component verdicts (for atlas calibration): market-as-epicenter: *Supported*. Market-as-spillover-site: *Contested-leaning supported* (direct evidence, withheld upstream data). Research-related incident: *Unresolved* (circumstantial conjunction, no direct evidence, uninspectable records). Deliberate engineering/release: *Contradicted* (IC consensus; not a bioweapon; genome lacks

engineering hallmarks per both literature and SAGO) ([FactCheck.org](#)) . Early-narrative-managed: *Supported* as to messaging discipline; *Contested* as to motive; *Contradicted* as to intelligence-community capture on the June 2026 documents ([senate.gov](#)) .

11. The epistemic environment

Public belief has been decoupled from the specialist literature for years: 64–66% of Americans leaned lab-leak in March 2023 polling while the two flagship *Science* papers had moved specialists toward the market; a 2021 PRRI survey found 39% of Americans believed the virus was “intentionally developed in a laboratory,” rising to 85% among QAnon believers — figures sociologists cite in mapping how the Fauci-centered conspiracy frame absorbed the legitimate lab-accident question ([The Washington Post](#)) . The Conversation’s analysis of the July 2026 hearing draws the line the atlas should keep bright: “asking whether SARS-CoV-2 emerged through either zoonotic spillover or a laboratory-related incident is not inherently a conspiracy theory,” but the transformation of “absence of evidence into evidence of concealment” is the signature move by which the question becomes one ([The Conversation](#)) .

The information environment degraded further from both directions in 2025–2026: a US government webpage asserting certainty the intelligence community does not hold, and a Chinese white paper reviving Fort Detrick insinuations while refusing the specific data requests (early sequences, animal sources, lab records) that would actually resolve the case ([FactCheck.org](#)) . Meanwhile the mortality ledger — roughly 14.8 million excess deaths across 2020–2021 by WHO estimate — is the standing argument for why the open questions in Sections 8–9 are worth what they cost ([nih.gov](#)) . The case is not closed; it is, in the literal sense, *unexamined* where it counts.

12. Condensed timeline (provenance table)

Date	Event	Key provenance
Apr 2012	Six Mojiang miners sickened	Master’ s thesis; Frontiers analysis
	clearing bat feces; three die	(Frontiers)
2013	WIV samples RaTG13 (96.1–96.3% match) at Tongguan mine	WIV records; RaTG13 history (wikipedia.org)
2014–2019	NIH→EcoHealth→WIV bat-CoV grant active (~\$120k/yr to WIV)	Grant records; Fauci testimony (InfluenceWatch)
Nov 2015	Baric-Shi chimeric spike paper (human airway infection)	Congressional recital (Rev)
27 Mar 2018	DEFUSE submitted to DARPA: FCS insertion, WIV partner, \$14.2M; rejected	Proposal text; Intercept (archive.org)
12 Sep 2019	WIV sequence database (~22k samples) goes offline	DRASTIC usage records; WaPo (The Washington Post)
Fall 2019	Several WIV researchers ill; symptoms “consistent but not diagnostic”	State Dept.; ODNI (The Washington Post)
~18 Nov 2019	Estimated lineage-B introduction (95% HPD 23 Oct–8 Dec)	Pekar et al. [(Science)]

Table 5 – continued

Date	Event	Key provenance
1 Dec 2019	Earliest known symptom onset (no market link found)	Lancet (The Lancet)
1 Jan 2020	Huanan market closed, cleared, disinfected	WHO/China timeline [(Science
31 Jan–1 Feb 2020	Andersen “inconsistent with evolutionary theory”; Farrar call; Fauci “tasks” email; diary: “not total agreement”	FOIA emails; diary release (USA Today)
4–17 Feb 2020	“Crackpot” email; rough draft to Fauci/Collins; Farrar “improbable” edit; preprint	Hearing record; MSK timeline (USA Today)
19 Feb 2020	Lancet letter condemns “conspiracy theories” (Daszak-organized, COI undeclared)	Lancet; THE (Times Higher Education (THE))
17 Mar 2020	“Proximal Origin” in Nature Medicine: no lab scenario “plausible”	Paper; hearing (The Intercept)
16–17 Apr 2020	Collins: “put down this very destructive conspiracy”; Fauci cites paper at podium	Hearing transcript (Rev)
Jan 2021	State Dept. fact sheet: sick WIV researchers, autumn 2019	WaPo op-ed record (The Washington Post)
Jun 2021	Bloom recovers 13 deleted early sequences from SRA	Science; BuzzFeed [(Science
Oct 2021	ODNI: 4 entities + NIC natural (low); FBI lab (moderate); 3 undecided	FactCheck summary (FactCheck.org)
26 Jul 2022	Worobey & Pekar Science papers: market epicenter; two introductions	Science [(Science
Feb–Mar 2023	DOE lab-lean (low); Wray public on FBI (moderate)	NPR; ABC (NPR)
23 Jun 2023	ODNI WIV report: BSL-2 work, no direct incident evidence	ODNI via HSDL; CBS (Homeland Security Digital Library)
8–9 Jan 2024	Fauci 14-hr interview: lab leak “not a conspiracy theory”; 6-ft “just appeared”	Committee memo (house.gov)
22 Jan 2024	Baric House interview: warned Shi on BSL-2; “load of BS”	Vanity Fair (Vanity Fair)
22 May 2024	Morens hearing: FOIA-evasion emails read into record	Committee wrap-up; KFF (house.gov)
3 Jun 2024	Fauci public hearing: denies prompting/editing; “open mind”	Rev transcript; Colorado Newslane (Rev)

Table 5 – continued

Date	Event	Key provenance
19 Sep 2024	Cell: wildlife DNA in all virus-positive samples from one stall	Cell; Cell Press (Cell Press)
2 Dec 2024	GOP 520-page report: lab “most likely”; DOJ grand jury revealed; minority dissents	Committee; Science; BMJ (house.gov)
17 Jan 2025	HHS debars EcoHealth & Daszak (5 yrs)	Committee; Kennedy (house.gov)
20 Jan 2025	Biden preemptive pardon (2014–Jan 19 2025)	CBS; The Hill (KARK)
25 Jan 2025	CIA: research-related origin more likely, low confidence	NBC (NBC News)
12 Mar 2025	BND 2020 assessment (80–95% lab accident) revealed; Merkel accused of burying	Telegraph; SZ via WSWS (The Telegraph)
18 Apr 2025	White House “Lab Leak: True Origins” page replaces covid.gov	NPR; ABC (NPR)
30 Apr 2025	China white paper: lab “extremely unlikely”; points at US/Fort Detrick	BBC Monitoring; SCIO (BBC Monitoring)
27 Jun 2025	WHO SAGO: weight of evidence zoonotic; lab not assessable; China withholding	WHO; report deck (WHO)
18–19 Jun 2026	ODNI “Fauci Files” (391 pp); Lawfare: documents don’t support headline claims	TNND; Lawfare (WBMA)
22 Jul 2026	Slack dump: Andersen 30%, Holmes 20%→10%; “handler”; “my spy”; FOIA curation	HSGAC; National Review (senate.gov)
23 Jul 2026	Baric 154-page Apr 2026 Senate interview released: DEFUSE “could have” made SARS-CoV-2-like virus	Washington Examiner (Washington Examiner)
~26–28 Jul 2026	1,000+ pages of Fauci diary (2019–2022) + 465-page prequel released	NPR (NPR)
29 Jul 2026	Fauci invokes Fifth Amendment 111 times under subpoena	NPR; PBS (NPR)
6 Aug 2026	HSGAC votes 8–5 contempt; referral sent to DOJ; DOJ decision pending	CBS; KARK (CBS News)

Prepared as an evidence seed for the Aletheia atlas ([ejhong.github.io/aletheia](#)). All claims carry source IDs preserved from retrieval; qualitative weights in Figures 2–3 are editorial syntheses to be replaced by

the atlas's own scoring. This document is for research/informational purposes and does not constitute legal, medical, or intelligence advice.

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